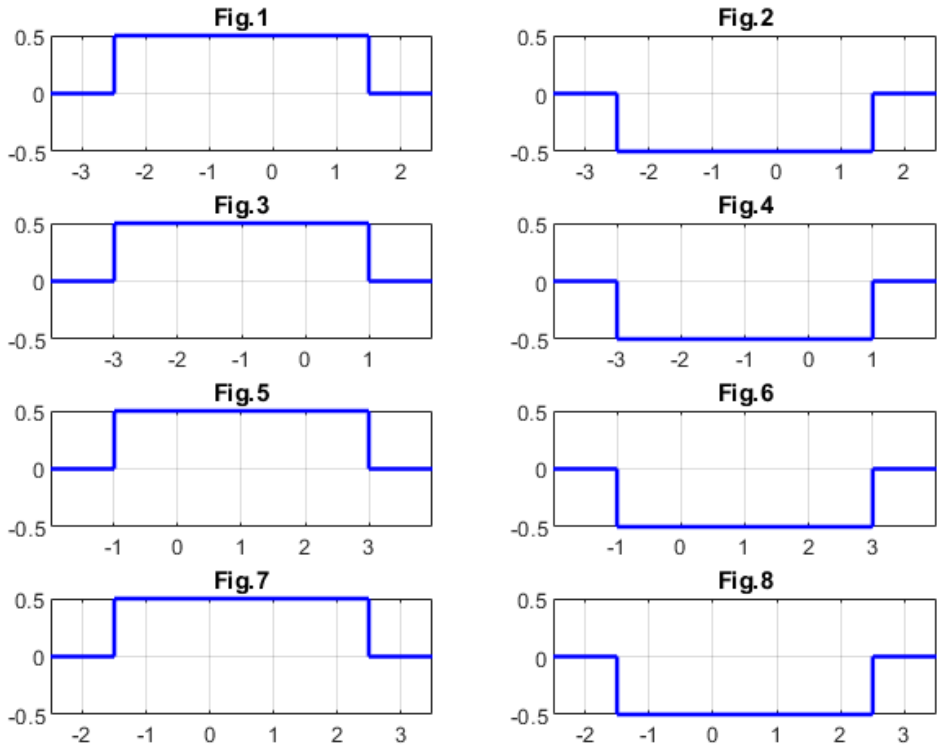
QUESTION 1: MULTIPLE CHOICE

10 out of 10 points

Which one of the below figures corresponds to the signal rect(t/4)*δ(-2t+1)?



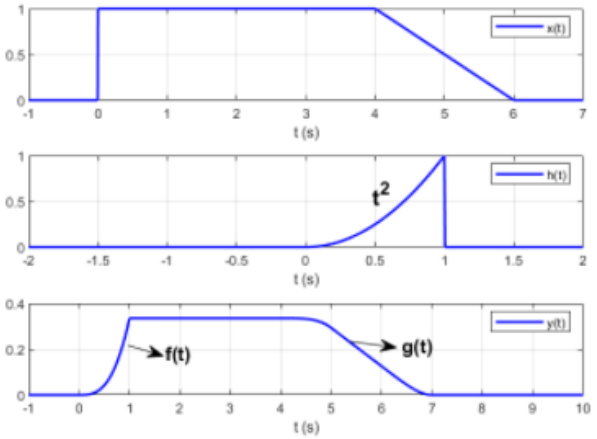
Given Answer: Fig. 7

Correct Answer: Fig. 7

QUESTION 2: CALCULATED NUMERIC

10 out of 10 points

Let y(t)=x(t)*h(t) as shown below:



What is the degree of the polynomial f(t)?

Given Answer: 3

Correct Answer: 3

Answer range +/- 0 (3 - 3)

QUESTION 3: MULTIPLE CHOICE

10 out of 10 points

A LTI system x(t) → y(t) is characterized by the following relation. What is the impulse response of the system?

$$y(t) = \int_{t+3}^{+\infty} x(t-3)\,dt$$

Given Answer: u(-t)

Correct Answer: u(-t)

QUESTION 4: TRUE/FALSE

5 out of 5 points

rect(t/2)δ(t-3)=0

Given Answer: True

Correct Answer: True

QUESTION 5: CALCULATED FORMULA

5 out of 5 points

If x(t)=comb(t/4), what is the period of x(t/4)?

Given Answer: 16

Correct Answer: 16

QUESTION 6: CALCULATED NUMERIC

10 out of 10 points

What is the value of the following integral?

$$\int_1^5 \mathrm{tri}\left(\frac{t}{5}\right)\delta\left(\frac{t}{2}-2\right)dt$$

Given Answer: 0.4

Correct Answer: 0.4

Answer range +/- 0 (0.4 - 0.4)

QUESTION 7: CALCULATED NUMERIC

10 out of 10 points

Let f(t)=3rect(t/2)*u(t).

If f(t)=at+b for -1≤t≤1, what is the value of the constant a?

Given Answer: 3

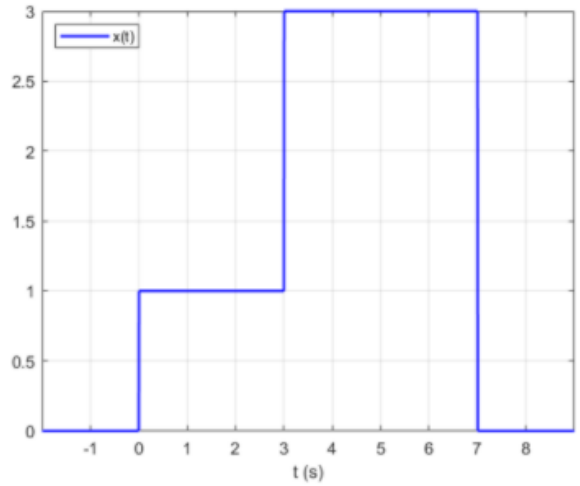
Correct Answer: 3

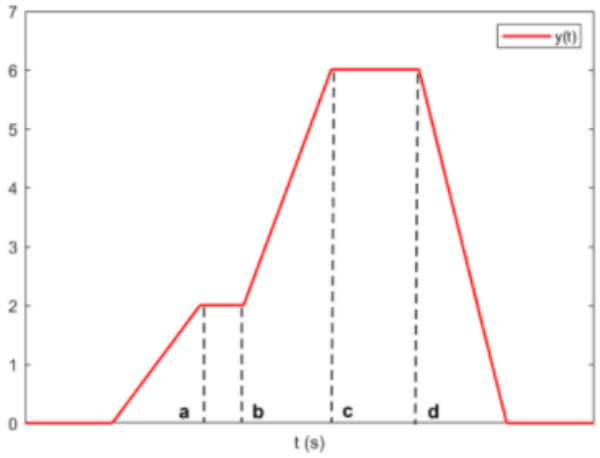
Answer range +/- 0 (3 - 3)

QUESTION 8: CALCULATED NUMERIC

10 out of 10 points

Consider the signal x(t) shown below. Let y(t)=rect(t/2)*x(t) where y(t) is plotted below. What is the value of the constant d?





Given Answer: 6

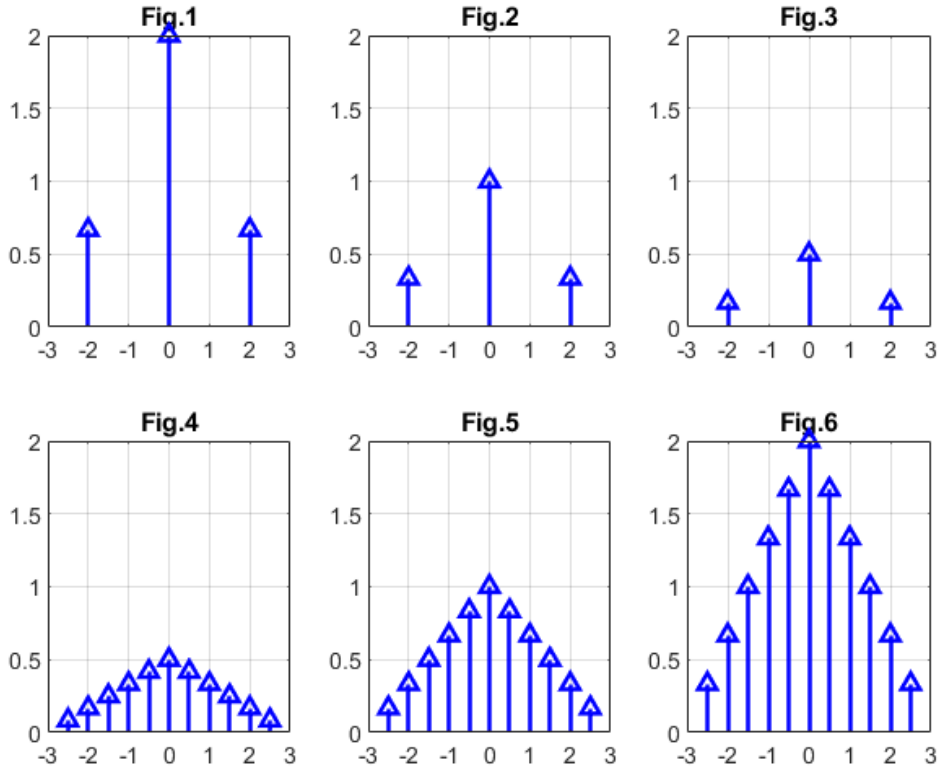
Correct Answer: 6

Answer range +/- 0 (6 - 6)

QUESTION 9: MULTIPLE CHOICE

10 out of 10 points

Which one of the below figures corresponds to the signal 4tri(t/3)comb(2t)?



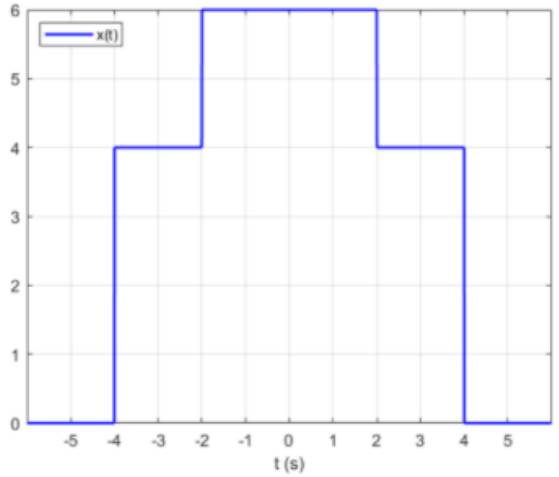
Given Answer: Fig. 6

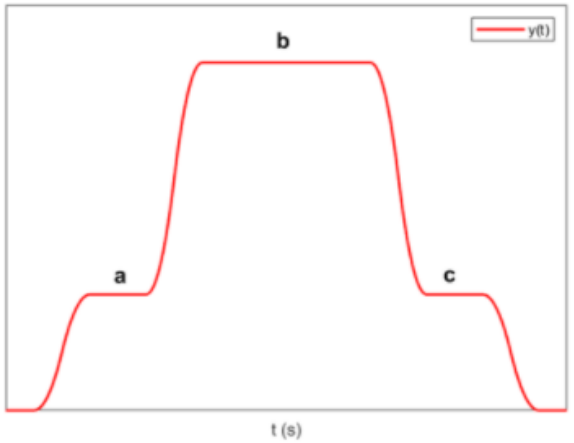
Correct Answer: Fig. 6

QUESTION 10: CALCULATED NUMERIC

10 out of 10 points

Consider the signal x(t) shown below. Let y(t)=tri(2t)*x(t) as plotted in the figure below. What is the value of the constant a?





Given Answer: 2

Correct Answer: 2

Answer range +/- 0 (2 - 2)

QUESTION 11: MULTIPLE CHOICE

10 out of 10 points

The convolution function of two signals is given by:

$$\begin{cases} 0 & t \le 0 \\ f(t) & 0 \le t \le 1 \\ 1/2 & t \ge 1 \end{cases}$$

What is the expression of f(t)?

Given Answer: $t-0.5t^2$

Correct Answer: $t-0.5t^2$